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DIELECTRIC FILTER

Abstract

A filtering device includes a multilayer body and a plurality of resonators. The multilayer body includes a plurality of dielectric layers and has a rectangular parallelepiped shape. Each of the plurality of resonators is located inside the multilayer body and extends in a first direction orthogonal to a stacking direction of the multilayer body. The multilayer body includes lateral surfaces in a direction orthogonal to the first direction, and includes a region where the plurality of resonators are arranged and regions where the plurality of resonators are not arranged. The first and second lateral surfaces in the former region extend to a greater extent in the first direction than the first and second lateral surfaces in the latter regions.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This is a continuation of International Application No. PCT/JP2023/030889 filed on Aug. 28, 2023 which claims priority from Japanese Patent Application No. 2022-189257 filed on Nov. 28, 2022. The contents of these applications are incorporated herein by reference in their entireties.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

[0002] The present disclosure relates to dielectric filters, and more specifically relates to a technique for preventing structural defects in a process of manufacturing a dielectric filter.

Description of the Related Art

[0003] Japanese Patent Laid-Open No. 2007-235465 discloses a bandpass filter with a multilayer dielectric resonator in which a plurality of internal electrode layers are stacked together inside a dielectric body. In the bandpass filter disclosed in Japanese Patent Laid-Open No. 2007-235465, an inductor portion of the internal electrode layer is formed in a longitudinal pattern, and shaped in such a manner that a part of the longitudinal pattern has a tapering width. Such a structural feature makes it possible to lower the resonance frequency without reducing the Q-factor, and accordingly, the resonator can be downsized.

BRIEF SUMMARY OF THE DISCLOSURE

[0004] The dielectric filter as disclosed in Japanese Patent Laid-Open No. 2007-235465 is used, for example, for filtering radio frequency signals for a small-sized mobile terminal which is typically a mobile phone or a smart phone.

[0005] Some dielectric filters are manufactured by stacking together a plurality of dielectric layers with plate conductors disposed thereon, and compression-bonding or sintering them. In a process of manufacturing such a multilayer dielectric filter, if the conductor density in the stacking direction is partially larger, a difference in thermal expansion coefficient between the portion where the conductor density is larger and a portion where the conductor density is smaller may cause structural defects including a crack between the conductor and the dielectric, and deformation of the internal structure such as bending or dimensional change of the conductor. Such structural defects may lead to device breakage or deterioration of the filtering characteristics.

[0006] The present disclosure is made to solve such problems, and an object of the present disclosure is to prevent structural defects in a process of manufacturing a dielectric filter.

[0007] A dielectric filter according to the present disclosure includes a multilayer body and a plurality of resonators. The multilayer body includes a plurality of dielectric layers and has a substantially rectangular parallelepiped shape. The plurality of resonators are located inside the multilayer body and extend in a first direction orthogonal to a stacking direction of the multilayer body. The multilayer body includes a first lateral surface and a second lateral surface in a direction orthogonal to the first direction, and includes a first region where the plurality of resonators are arranged and second regions where the plurality of resonators are not arranged. The first lateral surface and the second lateral surface in the first region extend to a greater extent in the first direction than the first lateral surface and the second lateral surface in the second regions.

[0008] The dielectric filter according to the present disclosure has the feature: the multilayer body defining the outer shape of the filter has the lateral surfaces that extend to a greater extent in the first region where the conductors of the resonators are arranged, than the lateral surfaces in the second regions where the conductors are not arranged. In a process of manufacturing the filter, such a feature enables reduction of stress between the first region and the second region caused by

shrinkage of the dielectric layers in the second region, to thereby enable prevention of structural defects of the dielectric filter.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0009] FIG. **1** is a block diagram of a communication apparatus including a radio frequency front-end circuit to which a filtering device according to Embodiment 1 is applicable.

[0010] FIG. **2** is an external perspective view of the filtering device according to Embodiment 1.

[0011] FIG. **3** is a transparent perspective view illustrating an internal structure of the filtering device according to Embodiment 1.

[0012] FIG. **4** is a side transparent view of the filtering device according to Embodiment 1, as seen in the positive direction of an X axis.

[0013] FIG. **5** is a side view of a filtering device according to Embodiment 2, as seen in the positive direction of a Y axis.

[0014] FIG. **6** is a cross-sectional view of the filtering device along line VI-VI in FIG. **5**.

[0015] FIG. **7** is a side view of a filtering device according to Embodiment 3, as seen in the positive direction of the Y axis.

[0016] FIG. **8** is a cross-sectional view of the filtering device along line VIII-VIII in FIG. **7**.

[0017] FIG. **9** is a cross-sectional view of the filtering device along line IX-IX in FIG. **7**.

[0018] FIG. **10** is a side view of a filtering device according to Embodiment 4, as seen in the positive direction of the Y axis.

[0019] FIG. **11** is a cross-sectional view of the filtering device along line XI-XI in FIG. **10**.

[0020] FIG. **12** is a cross-sectional view of the filtering device along line XII-XII in FIG. **10**.

[0021] FIG. **13** is a side view of a filtering device according to Embodiment 5, as seen in the positive direction of the Y axis.

[0022] FIG. **14** is a cross-sectional view of the filtering device along line XIV-XIV in FIG. **13**.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0023] Embodiments of the present disclosure are hereinafter described in detail with reference to the drawings. In the drawings, the same or corresponding parts are denoted by the same reference characters, and a description thereof is not herein repeated.

Embodiment 1

(Basic Configuration of Communication Apparatus)

[0024] FIG. **1** is a block diagram of a communication apparatus **10** including a radio frequency front-end circuit **20** to which a filtering device **100** of Embodiment 1 is applicable. Examples of communication apparatus **10** may include, for example, mobile terminals, typically smartphones, and base stations for mobile phones.

[0025] With reference to FIG. **1**, communication apparatus **10** includes an antenna **12**, a radio frequency front-end circuit **20**, a mixer **30**, a local oscillator **32**, a D/A converter (DAC) **40**, and an RF circuit **50**. Radio frequency front-end circuit **20** includes bandpass filters **22** and **28**, an amplifier **24**, and an attenuator **26**. In the following description referring to FIG. **1**, radio frequency front-end circuit **20** includes a transmission circuit that transmits radio frequency signals from antenna **12**, however, radio frequency front-end circuit **20** may also include a reception circuit that receives radio frequency signals through antenna **12**.

[0026] Communication apparatus **10** up-converts a signal transmitted from RF circuit **50** into a radio frequency signal and outputs the radio frequency signal from antenna **12**. A modulated digital signal output from RF circuit **50** is converted by D/A converter **40** into an analog signal. Mixer **30** mixes the analog signal generated by D/A converter **40** with an oscillation signal from local oscillator **32** to up-convert the resulting signal into a radio frequency signal. Bandpass filter **28**

removes any unwanted wave generated by the up-conversion and extracts only the signal within a desired frequency band. Attenuator **26** adjusts the intensity of the signal. Amplifier **24** amplifies the signal having passed through attenuator **26** to a predetermined power level. Bandpass filter **22** removes any unwanted wave generated during the amplification and passes only signal components within a frequency band specified by the communication standards. The signal having passed through bandpass filter **22** is emitted from antenna **12** as a transmission signal.

[0027] The filtering device as disclosed herein may be used as bandpass filters **22**, **28** of communication apparatus **10** as described above.

(Configuration of Filtering Device)

[0028] A configuration of a filtering device **100** according to Embodiment 1 is hereinafter described in detail with reference to FIGS. **2** to **4**. Filtering device **100** is a dielectric filter including a plurality of resonators, each of which defines and functions as a distributed parameter element.

[0029] FIG. **2** is an external perspective view of filtering device **100**. FIG. **2** shows only structural features of filtering device **100** that are visible from its outer surface side, and does not show its internal structural features. FIG. **3** is a transparent perspective view illustrating an internal structure of filtering device **100**. FIG. **4** is a side transparent view of filtering device **100**.

[0030] With reference to FIG. **2**, filtering device **100** includes a substantially rectangular parallelepiped multilayer body **110** in which a plurality of dielectric layers are stacked together in the stacking direction. In the description below, “Z-axis direction” refers to the stacking direction of multilayer body **110**, “X-axis direction” refers to the direction along the longer sides of multilayer body **110** and perpendicular to the Z-axis direction, and “Y-axis direction” (first direction) refers to the direction along the shorter sides of multilayer body **110**. In the description below, the upper side and the lower side may respectively refer to the positive direction of the Z axis and the negative direction of the Z axis in the drawings.

[0031] Multilayer body **110** has an upper surface **111**, a lower surface **112**, a lateral surface **113**, a lateral surface **114**, a lateral surface **115**, and a lateral surface **116**. Lateral surface **113** is a lateral surface of multilayer body **110** in the positive direction of the X axis, while lateral surface **114** is a lateral surface in the negative direction of the X axis. Lateral surfaces **115** and **116** are lateral surfaces of multilayer body **110** that are perpendicular to the Y-axis direction.

[0032] The dielectric layers of multilayer body **110** are each made of resin or ceramic such as low temperature co-fired ceramics (LTCC), for example. In multilayer body **110**, a plurality of plate conductors formed on respective dielectric layers and a plurality of vias formed between the dielectric layers provide the distributed parameter elements defining the resonators as well as capacitors and inductors to couple the distributed parameter elements. The “via” herein refers to a conductor that connects the respective conductors provided on different dielectric layers, and extends in the stacking direction. The via is formed of an electrically conductive paste, a plating, and/or a metallic pin, for example.

[0033] As illustrated in FIG. **2**, filtering device **100** includes shield conductors **121** and **122** that cover lateral surfaces **115** and **116**, respectively, of multilayer body **110**. Shield conductors **121** and **122** each also cover a part of upper surface **111** and a part of lower surface **112** of multilayer body **110**. As described in detail with reference to FIGS. **3** and **4**, lateral surfaces **115** and **116** of multilayer body **110** partially protrude in the Y-axis direction and accordingly shield conductors **121** and **122** also partially protrude in the Y-axis direction.

[0034] Respective portions of shield conductors **121** and **122** that are located on lower surface **112** of multilayer body **110** are connected, with connecting conductors such as solder bumps, to ground electrodes on a mounting substrate, which is not shown. Thus, shield conductors **121** and **122** also function as ground terminals.

[0035] Filtering device **100** also includes an input terminal **T1** and an output terminal **T2** on lower surface **112** of multilayer body **110**. Input terminal **T1** on lower surface **112** is located relatively closer to lateral surface **113** in the positive direction of the X axis. Output terminal **T2** on lower

surface **112** is located relatively closer to lateral surface **114** in the negative direction of the X axis. Input terminal **T1** and output terminal **T2** are connected, with connecting conductors such as solder bumps, to respective electrodes on the mounting substrate.

[0036] Next, with reference to FIGS. **3** and **4**, the internal structure of filtering device **100** is described. Filtering device **100** includes, in addition to the structural features illustrated in FIG. **2**, plate electrodes **130** and **135**, a plurality of resonators **141** to **145**, capacitor electrodes **161** to **165**, and connecting conductors **151** to **155** and **171** to **175**. In the following description, resonators **141** to **145**, capacitor electrodes **161** to **165**, and connecting conductors **151** to **155** and **171** to **175** may be referred to collectively as “resonator **140**,” “capacitor electrode **160**,” “connecting conductor **150**,” and “connecting conductor **170**,” respectively.

[0037] Plate electrodes **130** and **135** are arranged inside multilayer body **110** and located at respective positions spaced from each other and facing each other in the stacking direction (Z-axis direction). Plate electrode **130** is disposed on a dielectric layer close to upper surface **111**, and is connected to shield conductors **121** and **122** at respective ends along the X axis. As seen in a plan view in the stacking direction, plate electrode **130** has a shape that substantially covers the dielectric layers.

[0038] Plate electrode **135** is disposed on a dielectric layer close to lower surface **112** of multilayer body **110**. As seen in a plan view in the stacking direction, plate electrode **135** has a substantially H shape with cutouts formed in respective portions facing input terminal **T1** and output terminal **T2**. Plate electrode **135** is connected to shield conductors **121** and **122** at respective ends along the X axis.

[0039] In multilayer body **110**, resonators **141** to **145** are arranged between plate electrode **130** and plate electrode **135**. In filtering device **100**, resonators **141** to **145** are located inside multilayer body **110** and arranged in the X-axis direction. More specifically, resonators **141**, **142**, **143**, **144**, and **155** are arranged in this order from the positive direction toward the negative direction of the X axis.

[0040] Resonators **141** to **145** each extend in the Y-axis direction (first direction). Respective ends (first ends) of resonators **141** to **145** in the positive direction of the Y axis are connected to shield conductor **121**. Respective ends (second ends) of resonators **141** to **145** in the negative direction of the Y axis are spaced from shield conductor **122**.

[0041] Resonators **141** to **145** each include a plurality of conductors arranged in the stacking direction. The number of conductors defining each resonator is 13 or more, for example. In resonator **140**, a plurality of conductors defining each resonator are electrically connected together by connecting conductor **170**, at a position close to the second end on the shield conductor **122** side. Resonators **141** to **145** are connected to plate electrodes **130** and **135** through respective connecting conductors **151** to **155**, at respective positions close to the first end connected to shield conductor **121**. Each of connecting conductors **151** to **155** extends from plate electrode **130** to plate electrode **135** through a plurality of conductors of a respective one of the resonators. Each of connecting conductors **151** to **155** is electrically connected to a plurality of conductors defining a respective one of the resonators.

[0042] In such a configuration, most of electric current flowing through each resonator flows to ground terminals (i.e., plate electrodes **130** and **135** and shield conductor **121**) through a respective one of connecting conductors **151** to **155**. Therefore, the effective length of each resonator is the length from the second end to this connecting conductor. Each resonator is designed to have a length of $\lambda/4$ (FIG. **4**) from the second end to the connecting conductor (**151** to **155**). Resonator **140** functions as a distributed parameter TEM-mode resonator including a plurality of conductors as center conductors and plate electrodes **130** and **135** as outer conductors.

[0043] Resonator **141** is connected to input terminal **T1** through vias **V10** and **V11** and a plate electrode **PL1**. Resonator **145** is connected to output terminal **T2** through vias and a plate electrode **PL2**, which is not seen in FIG. **3** because they are hidden behind the resonator. Resonators **141** to

145 are magnetically coupled to one another, and a radio frequency signal input to input terminal **T1** is transmitted by resonators **141** to **145** in this order and output from output terminal **T2**. At this time, filtering device **100** functions as a bandpass filter, using the degree of coupling between the resonators.

[0044] The second-end side of each resonator **140** is provided with a respective one of capacitor electrodes **C10** to **C50** protruding towards an adjacent resonator. Some of a plurality of conductors of the resonator protrude outward to form the capacitor electrode. The degree of capacitive coupling between the resonators is adjustable by the length of the capacitor electrode in the Y-axis direction, the distance from the capacitor electrode to the adjacent resonator, and/or the number of conductors defining the capacitor electrode.

[0045] In filtering device **100**, capacitor electrode **C10** protrudes from resonator **141** toward resonator **142**, and capacitor electrode **C20** protrudes from resonator **142** toward resonator **141**, as illustrated in FIG. 3. Moreover, capacitor electrode **C30** protrudes from resonator **143** toward resonator **142**, and capacitor electrode **C40** protrudes from resonator **144** toward resonator **143**. Further, capacitor electrode **C50** protrudes from resonator **145** toward resonator **144**.

[0046] Capacitor electrodes **C10** to **C50** are not indispensable features and, as long as a desired degree of inter-resonator coupling is achievable, some or all of the capacitor electrodes may not be provided. In addition to the features illustrated in FIG. 3, the filtering device may further include a capacitor electrode protruding from resonator **142** toward resonator **143**, a capacitor electrode protruding from resonator **143** toward resonator **144**, and a capacitor electrode protruding from resonator **144** toward resonator **145**.

[0047] In addition, in filtering device **100**, capacitor electrodes **160** are arranged, facing the second ends of resonators **140**, respectively. The shape of each capacitor electrode **160** in a cross section parallel to a Z-X plane is similar to that of resonator **140**. Capacitor electrodes **160** are connected to shield conductor **122**. Thus, each resonator **140** and a respective one of capacitor electrodes **160** define a capacitor. The capacitance value of the capacitor defined by resonator **140** and respective capacitor electrode **160** is adjustable by adjusting a gap **GP** (distance in the Y-axis direction) between resonator **140** and capacitor electrode **160** illustrated in FIG. 4.

[0048] As described above with reference to FIG. 2, filtering device **100** of Embodiment 1 has lateral surfaces **115** and **116** of multilayer body **110** that partially protrude outward. More specifically, as illustrated in FIG. 4, lateral surfaces **115** and **116** of multilayer body **110** extend to a greater extent in the Y-axis direction in a region **RG1** where the conductors of resonator **140** and capacitor electrode **160** are arranged, than lateral surfaces **115** and **116** in a region **RG2** where the conductors of resonator **140** and capacitor electrode **160** are not arranged. In one example, region **RG1** is designed to have a dimension **h** in the stacking direction that is one-half or less of a dimension **H** in the stacking direction of multilayer body **110** ($h \leq H/2$). Region **R2** may include not only the region extending from region **RG1** toward upper surface **131** and the region extending from region **RG1** toward lower surface **132**, but also a region between the resonators and a region between the capacitor electrodes, and respective regions from the opposite endmost resonators to lateral surfaces **113** and **114** and respective regions from the opposite endmost capacitor electrodes to lateral surfaces **113** and **114**.

[0049] In filtering device **100** of Embodiment 1, each of shield conductors **121** and **122** has a double-layer structure formed by different electrical conductors. Specifically, shield conductor **121** includes two electrode layers **1211** and **1212**, and shield conductor **122** includes two electrode layers **1221** and **1222**.

[0050] Electrode layers **1211** and **1221** are formed by applying or printing, on the surface of multilayer body **110**, an electrically conductive paste containing copper (Cu), nickel (Ni), silver (Ag), or the like, and firing the paste to thereby solidify the paste. In the following, an electrode layer formed by such printing or the like is also referred to as “underlying electrode.” Electrode layers **1212** and **1222** are formed by performing sputtering or plating of nickel, tin (Sn), or Ni—Sn

alloy on electrode layers **1211** and **1221** serving as underlying electrodes. Electrode layers **1211** and **1221** are greater in thickness than electrode layers **1212** and **1222**.

[0051] Shield conductors **121** and **122** each cover a part of upper surface **111** and a part of lower surface **112**, and cover lateral surfaces **115** and **116**, respectively. Accordingly, shield conductors **121** and **122** in regions RG1 of lateral surfaces **115** and **116** protrude outward relative to shield conductors **121** and **122** in regions RG2.

[0052] The multilayer dielectric filter as described above is generally manufactured by stacking together a plurality of dielectric layers with plate conductors disposed thereon, and compression-bonding or sintering them. The shrinkage of the dielectric such as ceramic is larger than the shrinkage of the conductor, and therefore, in a process of manufacturing the dielectric filter, if there are dielectric layers with a high conductor density in the stacking direction like region RG1, and dielectric layers with a low conductor density like region RG2, a difference in thermal expansion coefficient between these two regions causes stress applied to the conductor in the direction of compression, at the interface between the conductor and the dielectric. As a result, structural defects may occur between the conductor and the dielectric, such as a crack, separation between dielectric layers, deformation due to buckling of the conductor, and/or deterioration of the flatness of the surface of the multilayer body. Occurrence of such structural defects may cause reduction of the strength of the filtering device, or shortening of the device life. Further, there is a possibility that a capacitance value and an inductance value intended by design cannot be implemented, resulting in deterioration of the filtering characteristics.

[0053] Filtering device **100** of Embodiment 1 has the feature: dielectric layers of region RG1 where conductors of multilayer body **110** are arranged protrude relative to dielectric layers of region RG2 where the conductors are not arranged. Such a feature can be achieved by adjusting the temperature increase profile in the firing step and making the timing at which the dielectric (ceramic) in region RG2 shrinks different from the timing at which the conductor in region RG1 shrinks. Such adjustment of the shrinkage timing reduces the stress generated at the interface between the dielectric and the conductor, so that occurrence of structural defects such as crack can be prevented in the process of manufacturing filtering device **100**. Accordingly, breakage, for example, of filtering device **100** can be prevented, and deterioration of the filtering characteristics can be suppressed.

[0054] Region RG2 that is sufficiently larger than region RG1 makes it easier to alleviate the stress occurring to the conductor during shrinkage in region RG1. It is therefore preferable to set dimension h of region RG1 in the stacking direction to one-half or less of dimension H of multilayer body **110** in the stacking direction.

[0055] Moreover, shield conductors **121** and **122** can have steps formed so as to conform to the protrusion. Such a structure can prevent the solder from extending toward upper surface **111**, when filtering device **100** is mounted on a substrate by soldering.

[0056] “Lateral surface **115**” and “lateral surface **116**” in Embodiment 1 correspond to “first lateral surface” and “second lateral surface” in the present disclosure, respectively. “Region RG1” and “region RG2” in Embodiment 1 correspond to “first region” and “second region” in the present disclosure, respectively. “Plate electrode **130**” and “plate electrode **135**” in Embodiment 1 correspond to “first plate electrode” and “second plate electrode” in the present disclosure, respectively. “Shield conductor **121**” and “Shield conductor **122**” in Embodiment 1 correspond to “first shield conductor” and “second shield conductor” in the present disclosure, respectively. Each of “connecting conductors **151** to **155**” in Embodiment 1 corresponds to “first connecting conductor” in the present disclosure. Each of “connecting conductors **171** to **175**” in Embodiment 1 corresponds to “second connecting conductor” in the present disclosure.

Embodiment 2

[0057] In connection with Embodiment 2, another configuration of the shield conductors formed on lateral surfaces **115** and **116** of multilayer body **110** is described.

[0058] FIG. 5 is a side view of a filtering device **100A** according to Embodiment 2, as seen in the positive direction of the Y axis. FIG. 6 is a cross-sectional view of filtering device **100A** along line VI-VI in FIG. 5.

[0059] Shield conductors **121** and **122** of filtering device **100** in Embodiment 1 are disposed to cover the whole of lateral surface **115** and the whole of lateral surface **116**, respectively. In contrast, as illustrated in FIG. 5, in the portion where resonator **140** is formed, a shield conductor **121A** disposed on lateral surface **115** of filtering device **100A** in Embodiment 2 has a cutout **125** formed in each of respective regions RG2 on the upper side and the lower side of resonator **140**. Thus, in these regions, lateral surface **115** of multilayer body **110** is exposed.

[0060] As illustrated in FIG. 6, in the cross section of the portion where cutouts **125** are formed, shield conductors **121A** and **122A** are disposed on only respective ends of resonator **140** and capacitor electrode **160**. In contrast, in the portion between adjacent resonators and respective portions between the endmost resonators and lateral surfaces **113** and **114**, shield conductors **121A** and **122A** are disposed from upper surface **111** through lateral surface **115**, **116** to lower surface **112**. The shield conductor located in the portions where the resonator is not disposed ensures connection with plate electrodes **130** and **135** in multilayer body **110**.

[0061] If, like filtering device **100**, the firing step is performed in the state where the whole of lateral surfaces **115** and **116** is covered with electrode layers **1211** and **1221** which are underlying electrodes of shield conductors **121** and **122**, the dielectric restrained by electrode layers **1211** and **1221** may cause the cross section of the multilayer body to be deformed into a drum shape. As a result, tensile stress in the stacking direction may be applied to the dielectric layers, to cause separation between dielectric layers and/or separation between the dielectric and the conductor. Like filtering device **100A** of Embodiment 2, in the portion of lateral surface **115** where resonator **140** is disposed, cutouts **125** formed on the upper side and the lower side of resonator **140** alleviate restriction of region RG2 caused by the shield conductors. Thus, it is possible to reduce occurrence of stress in the stacking direction and prevent structural defects due to separation between the dielectric and the conductor.

[0062] Moreover, the difference in shield conductor shrinkage between region RG1 and region RG2 is eliminated, and therefore, it is also possible to reduce nonlinear distortion of the end of the conductor due to the difference in shield conductor shrinkage.

Embodiment 3

[0063] In connection with Embodiment 3, a configuration is described where the shield conductor in region RG1 with conductors is formed by sputtering or plating only, without using the underlying electrode.

[0064] FIG. 7 is a side view of a filtering device **100B** according to Embodiment 3, as seen in the positive direction of the Y axis. FIG. 8 is a cross-sectional view of filtering device **100B** along line VIII-VIII of FIG. 7. FIG. 9 is a cross-sectional view of filtering device **100B** along line IX-IX in FIG. 7.

[0065] With reference to FIGS. 7 to 9, like filtering device **100A** of Embodiment 2, filtering device **100B** according to Embodiment 3 has cutouts **125** formed in shield conductors **121B** and **122B** and located on the upper side and the lower side of resonator **140** and capacitor electrode **160**.

However, as illustrated in FIGS. 8 and 9, electrode layers **121B1** and **122B1** of the underlying electrodes are not disposed on respective ends of the conductors of resonator **140** and capacitor electrode **160** in region RG1, and only electrode layers **121B2** and **122B2** formed by sputtering or plating are disposed. Namely, electrode layers **121B1** and **122B1** are disposed between adjacent resonators **140**, between adjacent capacitor electrodes **160**, and respective portions where lateral surfaces **115** and **116** are contiguous to lateral surfaces **113** and **114**.

[0066] As described above, when electrode layers **121B1** and **122B1** serving as underlying electrodes are applied and fired, stress is generated due to restraint of the dielectric layers by electrode layers **121B1** and **122B1**. Therefore, the underlying electrodes are not provided in the

region where the conductors of resonator **140** and capacitor electrode **160** are disposed, to thereby enable reduction of the stress in this region during the firing step. Thus, the stress acting on the dielectric and the conductor can be suppressed, so that occurrence of structural defects in the process of manufacturing can be prevented.

[0067] In filtering device **100B**, glass containing a metal component (e.g., copper) that easily diffuses in the firing step is added to the dielectric in the portion (region **RG1**) where the conductors of resonator **140** and capacitor electrode **160** are disposed. Thus, the concentration of copper contained in the dielectric layers can be made higher than that in region **RG2**, to improve the adhesion of the metal member adhered by sputtering and plating.

[0068] In addition, in filtering device **100B** as illustrated in FIG. **9**, the end of the conductor of resonator **140** in lateral surface **115** and the end of the conductor of capacitor electrode **160** in lateral surface **116** each have a larger conductor width than the conductor width of the other portions. The enlarged conductor portions are connected respectively to electrode layers **121B1** and **122B1** serving as underlying electrodes. This ensures electrical connection between the conductors of resonator **140** and capacitor electrode **160**, and plate electrodes **130** and **135**.

[0069] Further, each of capacitor electrodes **160** is provided with a connecting conductor **180** that connects the conductors of capacitor electrode **160** and plate electrodes **130** and **135**. Thus, reliable electrical connection between capacitor electrode **160** and plate electrodes **130** and **135** can be achieved.

[0070] “Connecting conductor **180**” in Embodiment 3 corresponds to “third connecting conductor” in the present disclosure.

Embodiment 4

[0071] In Embodiment 3, only the shield conductors in the portions where the conductors of resonator **140** and capacitor electrode **160** are disposed are formed by sputtering or plating, and the shield conductors in the other portions have a double-layer structure in which underlying electrodes are used.

[0072] In connection with Embodiment 4, a configuration where shield conductors disposed on lateral surfaces **115** and **116** are entirely formed by sputtering or plating is described.

[0073] FIG. **10** is a side view of a filtering device **100C** according to Embodiment 4, as seen in the positive direction of the Y axis. FIG. **11** is a cross-sectional view of filtering device **100C** along line XI-XI in FIG. **10**. FIG. **12** is a cross-sectional view of filtering device **100C** along line XII-XII in FIG. **10**.

[0074] Referring to FIGS. **10** to **12**, shield conductors **121C** and **122C** of filtering device **100C** in Embodiment 4 are basically shaped identically to those of filtering device **100A** in Embodiment 2, and cutouts **125** are formed on the upper side and the lower side of resonator **140**. However, as illustrated in FIGS. **11** and **12**, shield conductors **121C** and **122C** are conductors having a single-layer structure formed by sputtering or plating. Thus, the shield conductors can be formed by only sputtering or plating without using underlying electrodes, to eliminate the restraint of the dielectric layers by the underlying electrodes in the firing step. It is therefore possible to suppress stress between the dielectric and the conductor, and prevent occurrence of structural defects in the process of manufacturing.

[0075] In the case where plating is performed, it is desirable to increase the electrical conductivity of the region where the shield conductor is to be formed. In view of this, in filtering device **100C** as illustrated FIG. **12**, resonators adjacent to each other and capacitor electrodes adjacent to each other are connected to each other in the proximity of the lateral surfaces. Further, in filtering device **100C**, a plurality of plate electrodes **190** are stacked in the vicinity of lateral surfaces **115** and **116**, in region **RG2** where cutouts **125** are not formed. The ends of plate electrodes **190** are exposed to lateral surfaces **115** and **116**. Thus, the electrical conductivity of lateral surfaces **115** and **116** in this region can be made substantially identical to that of the resonator portion, so that the adhesion of the metal member in the plating process can be enhanced.

[0076] Like Embodiment 3, in filtering device **100C**, glass containing a metal component easily diffused in the firing step may be added to the dielectric layers in region **RG1** and the dielectric layers in regions **RG2** where plate electrodes **190** are disposed.

[0077] In this way, the shield conductors on the lateral surfaces of the multilayer body are formed by sputtering or plating without using underlying electrodes, to thereby enable reduction of the stress generated in the firing step, and prevention of occurrence of structural defects.

Embodiment 5

[0078] In connection with Embodiment 5, a configuration is described where shield conductors formed by sputtering or plating are disposed from the upper surface to the lower surface, only on the portions of the lateral surfaces where the resonators and the capacitor electrodes are located.

[0079] FIG. **13** is a side view of a filtering device **100D** according to Embodiment 5, as seen in the positive direction of the Y axis. FIG. **14** is a cross-sectional view of filtering device **100D** along line XIV-XIV in FIG. **13**.

[0080] In filtering device **100D** as illustrated in FIG. **13**, a shield conductor **121D** is disposed from upper surface **111** to lower surface **112** through lateral surface **115**, only in the portions where resonators **140** are disposed. Namely, on lateral surface **115**, the shield conductor is not disposed in the region between resonators adjacent to each other and the region close to each of lateral surfaces **113** and **114**.

[0081] Shield conductor **121D** in this region of the lateral surface is formed by sputtering or plating without using the underlying electrode. Therefore, a plurality of plate electrodes **195** are stacked in the vicinity of lateral surface **115**, in regions **RG2** on the upper side and the lower side of resonator **140** in multilayer body **110**. This plate electrode **195** can enhance the adhesion of the metal member in the sputtering or plating process.

[0082] On lateral surface **116** as well, a shield conductor **122D** is disposed to extend from upper surface **111** to lower surface **112** through lateral surface **116**, only in the region where capacitor electrode **160** is disposed. A plurality of plate electrodes **196** are stacked in the vicinity of lateral surface **116**, in regions **RG2** on the upper side and the lower side of capacitor electrode **160** in multilayer body **110**. This plate electrode **196** can enhance the adhesion of the metal member to lateral surface **116** in the sputtering or plating process.

[0083] Shield conductors **121D** and **122D** on upper surface **111** and lower surface **112** are formed in a double-layer structure made up of an electrode (underlying electrode) formed by a process such as printing and an electrode formed by sputtering or plating. Specifically, shield conductor **121D** includes an electrode layer **1211D** serving as the underlying electrode and an electrode layer **1212D** formed by plating for example. Shield conductor **122D** includes an electrode layer **1221D** serving as the underlying electrode and an electrode layer **1222D** formed by plating, for example.

[0084] On upper surface **111** and lower surface **112**, it is necessary to ensure more reliable electrical connection between connecting conductor **150** for resonator **140** and shield conductor **121D**, and electrical connection between connecting conductor **180** for capacitor electrode **160** and shield conductor **122D**. In view of this, in filtering device **100D**, the double-layer structure using the underlying electrode is adopted for shield conductors **121D** and **122D** on upper surface **111** and lower surface **112**. Since the underlying electrode is disposed on only upper surface **111** and lower surface **112**, stress due to the electrode between the dielectric and the conductor hardly occurs in the firing step.

[0085] Like Embodiment 3, in filtering device **100D**, glass containing a metal component easily diffused in the firing step may be added to the dielectric layers in region **RG1** and the dielectric layers in regions **RG2** where plate electrodes **195** and **196** are disposed.

[0086] In this way, the shield conductors on the lateral surfaces of the multilayer body are formed by sputtering or plating without using underlying electrodes, to thereby enable reduction of the stress generated in the firing step, and prevention of occurrence of structural defects.

(Aspects)

[0087] It should be understood by those skilled in the art that the foregoing embodiments are specific examples of the following aspects.

[0088] (Clause 1) A dielectric filter according to one aspect includes a multilayer body and a plurality of resonators. The multilayer body includes a plurality of dielectric layers and has a substantially rectangular parallelepiped shape. The plurality of resonators are located inside the multilayer body and extend in a first direction orthogonal to a stacking direction of the multilayer body. The multilayer body includes a first lateral surface and a second lateral surface in a direction orthogonal to the first direction, and includes a first region where the plurality of resonators are arranged and second regions where the plurality of resonators are not arranged. The first lateral surface and the second lateral surface in the first region extend to a greater extent in the first direction than the first lateral surface and the second lateral surface in the second regions.

[0089] (Clause 2) The dielectric filter according to Clause 1 further includes a first plate electrode and a second plate electrode, and a first shield conductor and a second shield conductor. The first plate electrode and the second plate electrode are located inside the multilayer body and spaced from each other in the stacking direction. The first shield conductor and a second shield conductor are arranged respectively on the first lateral surface and the second lateral surface of the multilayer body, and connected to the first plate electrode and the second plate electrode. The plurality of resonators are arranged between the first plate electrode and the second plate electrode. The plurality of resonators each have a first end connected to the first shield conductor and a second end spaced from the second shield conductor.

[0090] (Clause 3) In the dielectric filter according to Clause 2, in the first shield conductor, a cutout is formed in at least a part of each of the second regions that extend respectively toward the first plate electrode and the second plate electrode from the first region.

[0091] (Clause 4) In the dielectric filter according to Clause 2, each of the plurality of resonators includes a plurality of conductors extending in the first direction and stacked together in the stacking direction.

[0092] (Clause 5) The dielectric filter according to Clause 4 further includes a first connecting conductor disposed at the first end of each resonator of the plurality of resonators, connecting the resonator to the first plate electrode and the second plate electrode, and electrically connecting the plurality of conductors to each other.

[0093] (Clause 6) The dielectric filter according to Clause 4 or 5 further includes a second connecting conductor disposed at the second end of each resonator of the plurality of resonators, and electrically connecting the plurality of conductors to each other.

[0094] (Clause 7) The dielectric filter according to Clause 2 further includes a capacitor electrode facing the second end of each resonator of the plurality of resonators, and connected to the second shield conductor.

[0095] (Clause 8) In the dielectric filter according to Clause 7, in the second shield conductor, a cutout is formed in at least a part of each of the second regions that extend respectively toward the first plate electrode and the second plate electrode from the first region.

[0096] (Clause 9) In the dielectric filter according to Clause 7, the capacitor electrode includes a plurality of conductors extending in the first direction and stacked together in the stacking direction. The dielectric filter further includes a third connecting conductor connecting the capacitor electrode to the first plate electrode and the second plate electrode, and electrically connecting the plurality of conductors to each other.

[0097] (Clause 10) In the dielectric filter according to Clause 2, the first shield conductor and the second shield conductor are formed by applying a metal paste to the first lateral surface and the second lateral surface respectively and firing the metal paste.

[0098] (Clause 11) In the dielectric filter according to Clause 2, the first shield conductor and the second shield conductor are formed by performing sputtering or plating on the first lateral surface and the second lateral surface, respectively.

[0099] (Clause 12) In the dielectric filter according to Clause 11, a concentration of copper included in the dielectric layers in the first region is higher than a concentration of copper included in the dielectric layers in the second regions.

[0100] (Clause 13) In the dielectric filter according to Clause 2, the first shield conductor and the second shield conductor are formed by applying a metal paste to the first lateral surface and the second lateral surface respectively and firing the metal paste, and thereafter performing sputtering or plating.

[0101] (Clause 14) In the dielectric filter according to Clause 1, in the stacking direction, a dimension of the first region is one-half or less of a dimension of the multilayer body.

[0102] It should be construed that the embodiments disclosed herein are given by way of illustration in all respects, not by way of limitation. It is intended that the scope of the present disclosure is defined by claims, not by the above description of the embodiments, and encompasses all modifications and variations equivalent in meaning and scope to the claims.

[0103] **10** communication apparatus; **12** antenna; **20** radio frequency front-end circuit; **22, 28** bandpass filter; **24** amplifier; **26** attenuator; **30** mixer; **32** local oscillator; **40** D/A converter; **50** RF circuit; **100, 100A-100D** filtering device; **110** multilayer body; **111** upper surface; **112** lower surface; **113-116** lateral surface; **121, 121A-121D, 122, 122A-122D** shield conductor; **125** cutout; **130, 135, 190, 195, 196, PL1, PL2** plate electrode; **140-145** resonator; **150-155, 170-175, 180** connecting conductor; **160, 161-165, C10-C50** capacitor electrode; **121B1, 121B2, 122B1, 122B2, 1211, 1212, 1221, 1222, 1211D, 1212D, 1221D, 1222D** electrode layer; **RG1, RG2** region; **T1** input terminal; **T2** output terminal; **V10, V11** via.

Claims

1. A dielectric filter comprising: a multilayer body including a plurality of dielectric layers and having a substantially rectangular parallelepiped shape; and a plurality of resonators located inside the multilayer body and extending in a first direction orthogonal to a stacking direction of the multilayer body; wherein the multilayer body includes a first lateral surface and a second lateral surface in a direction orthogonal to the first direction, and includes a first region where the plurality of resonators are arranged and second regions where the plurality of resonators are not arranged, and the first lateral surface and the second lateral surface in the first region extend to a greater extent in the first direction than the first lateral surface and the second lateral surface in the second regions.
2. The dielectric filter according to claim 1, further comprising: a first plate electrode and a second plate electrode located inside the multilayer body and spaced from each other in the stacking direction; and a first shield conductor and a second shield conductor arranged respectively on the first lateral surface and the second lateral surface of the multilayer body, and connected to the first plate electrode and the second plate electrode, wherein the plurality of resonators are arranged between the first plate electrode and the second plate electrode, and the plurality of resonators each have a first end connected to the first shield conductor and a second end spaced from the second shield conductor.
3. The dielectric filter according to claim 2, wherein in the first shield conductor, a cutout is provided in at least a part of each of the second regions extending respectively toward the first plate electrode and the second plate electrode from the first region.
4. The dielectric filter according to claim 2, wherein each of the plurality of resonators includes a plurality of conductors extending in the first direction and stacked together in the stacking direction.
5. The dielectric filter according to claim 4, further comprising a first connecting conductor disposed at the first end of each resonator of the plurality of resonators, connecting the resonator to the first plate electrode and the second plate electrode, and electrically connecting the plurality of

conductors to each other.

6. The dielectric filter according to claim 4, further comprising a second connecting conductor disposed at the second end of each resonator of the plurality of resonators, and electrically connecting the plurality of conductors to each other.

7. The dielectric filter according to claim 2, further comprising a capacitor electrode facing the second end of each resonator of the plurality of resonators and connected to the second shield conductor.

8. The dielectric filter according to claim 7, wherein in the second shield conductor, a cutout is provided in at least a part of each of the second regions extending respectively toward the first plate electrode and the second plate electrode from the first region.

9. The dielectric filter according to claim 7, wherein the capacitor electrode includes a plurality of conductors extending in the first direction and stacked together in the stacking direction, and the dielectric filter further comprises a third connecting conductor connecting the capacitor electrode to the first plate electrode and the second plate electrode and electrically connecting the plurality of conductors to each other.

10. The dielectric filter according to claim 2, wherein the first shield conductor and the second shield conductor are provided by applying a metal paste to the first lateral surface and the second lateral surface respectively and firing the metal paste.

11. The dielectric filter according to claim 2, wherein the first shield conductor and the second shield conductor are provided by performing sputtering or plating on the first lateral surface and the second lateral surface, respectively.

12. The dielectric filter according to claim 11, wherein a concentration of copper included in the dielectric layers in the first region is higher than a concentration of copper included in the dielectric layers in the second regions.

13. The dielectric filter according to claim 2, wherein the first shield conductor and the second shield conductor are provided by applying a metal paste to the first lateral surface and the second lateral surface respectively and firing the metal paste, and thereafter performing sputtering or plating.

14. The dielectric filter according to claim 1, wherein, in the stacking direction, a dimension of the first region is one-half or less of a dimension of the multilayer body.

15. The dielectric filter according to claim 5, further comprising a second connecting conductor disposed at the second end of each resonator of the plurality of resonators, and electrically connecting the plurality of conductors to each other.
